$$2 = \left(\frac{1}{2}\right)^{n-1}$$

$$2 = \left(2^{-1}\right)^{n-1}$$

$$2^1 = \left(2^{-n+1}\right)$$

$$n = 0$$

21. Answer : option(a)

	A	$\longrightarrow$	B	C	D
Initial	a		0	0	0
Reacted at time t	x		–	–	–
After time t	$(a-x)$		x	x	x
Total number of moles		$=(a+2x)$			

$$a \propto P_0$$

$$(a+2x) \propto P$$

$$\frac{a}{(a+2x)} = \frac{P_0}{P}$$

$$x = \frac{(P-P_0)a}{2P_0}$$

$$(a-x) = a - \left(\frac{(P-P_0)a}{2P_0}\right)$$

$$(a-x) = a\left\{\frac{3P_0-P}{2P_0}\right\}$$

$$k = \left(\frac{2.303}{t}\right)\log\frac{[A_0]}{[A]}$$

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{a}{a-x}\right)$$

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{a}{a\left\{\frac{3P_0-P}{2P_0}\right\}}\right)$$

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{2P_0}{3P_0-P}\right)$$

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